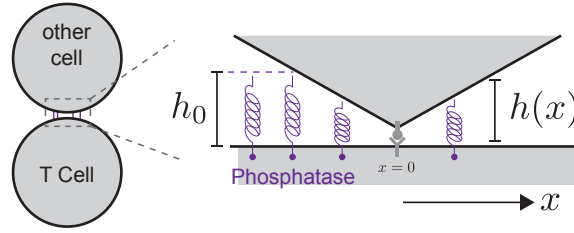


Problem Set 5: Kinetic segregation

Physics 230A/146A — Biological Physics

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T Cells have molecules on their surface called phosphatases. These phosphatases have large bulky extracellular domains. So, when another cell pushes up against the T Cell, the phosphatases get squeezed. Phosphatases are more squeezed near a receptor, so they tend to move away from the receptor. This phenomenon is called *kinetic segregation*.



Suppose there is a receptor at $x = 0$. For simplicity, let us study the problem in 1-dimension, and assume the height of the other cell's membrane at location x is given by

$$h(x) = \alpha|x| = \begin{cases} -\alpha x & \text{if } x < 0 \\ +\alpha x & \text{if } x > 0 \end{cases} \quad (1)$$

Assume the phosphatase molecules are vertical springs, with rest-length h_0 and spring constant k , so each one has an energy

$$E = \frac{1}{2}k(h - h_0)^2 \quad \text{if } h < h_0.$$

and $E = 0$ if $h > h_0$.

- i. As the phosphatase moves to the right away from the receptor, it becomes less and less compressed. At what x -coordinate will the phosphatase molecules not be compressed at all? Call this x_{far} .

- ii. If a phosphatase molecule is at coordinate x , then given the membrane has height $h(x)$ from Equation 1, what is its energy $E(x)$?

Hint: There will be 4 cases:

$$\begin{aligned} x &< -x_{\text{far}} \\ -x_{\text{far}} &< x < 0 \\ 0 &< x < x_{\text{far}} \\ x_{\text{far}} &< x \end{aligned}$$

- iii. At steady state, what is the concentration profile of the phosphatase $c(x)$?

Assume that the phosphatase undergo diffusion and advection according to

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} + \mu \frac{\partial}{\partial x} \left(c \frac{\partial E}{\partial x} \right),$$

which means that the steady-state concentration profile satisfies

$$c(x) = c_0 \exp \left(-\frac{\mu}{D} E(x) \right)$$

where c_0 is an unspecified constant.

- iv. What is the concentration at the receptor, $c(0)$, compared to the concentration far from the receptor, $c(x_{\text{far}})$? Express it as a ratio. Is it less than half?

You may assume $D/\mu = 4 \text{ pN nm}$, $h_0 = 20 \text{ nm}$, $k = 0.1 \text{ pN/nm}$, and $\alpha = 0.1$.